

Instructions

1. Run **Task 0** in the notebook first. Every other task depends on it.
2. Work through Tasks 1 to 5 in order.
3. Write your code in the empty cells provided beneath each question.
4. If a cell produces an error, read the error message — it usually explains the fix.
5. Save your notebook regularly with **Ctrl+S**.
6. Before submitting, run **Kernel > Restart & Run All** to confirm clean execution.

Files provided with this exam

unseen_university.db	SQLite database — open with <code>sqlite3.connect('unseen_university.db')</code>	Tasks 0 – 4
late_enrolments.csv	CSV of new students from Überwald — load with <code>pd.read_csv('late_enrolments.csv')</code>	

Scenario

Welcome to Unseen University, Ankh-Morpork's premier institution of magical learning. After centuries of record-keeping by increasingly confused wizards (and one orangutan), the University has decided to modernise its student registry.

You have been appointed as the new **Database Registrar**. The database contains four tables:

teachers	teacher_id, name, dob, email
programs	program_id, program_name, program_lead → teachers(teacher_id)
modules	module_id, module_name, teacher_id → teachers(teacher_id), duration_hrs
students	student_id, name, dob, gender, program_id → programs(program_id)

Notice that no table contains a list. Relationships between tables are handled using **foreign keys** and **queries** — exactly as discussed in class.

Mark Breakdown

Task	Description	Marks
1	Database Exploration	10
2	Queries	15
3	Data Entry Form	10
4	Report and Visualisation	10
5	Import External Data	5
Total		50

Authorship Declaration

I hereby declare that all work described in this submission, except where referenced, is entirely my own. All sources of information have been acknowledged.

Name (print):

Date:

Signed:

Task 0 — Setup — Run This Cell First

marks

The first code cell in the notebook creates the database and loads all sample data. Run it before attempting any other task. You do not need to modify it.

Task 1 — Database Exploration

10 marks

The Archchancellor has demanded a full audit of the registry before the new term begins.

1a) View the structure of the students table

(3 marks)

Display the column names and data types for the **students** table using `PRAGMA table_info`.

1b) Count the enrolled students

(3 marks)

Using SQL, find out how many students are currently enrolled at Unseen University.

1c) Preview the modules table

(4 marks)

Load the **modules** table into a pandas DataFrame and display the first 5 rows.

Task 2 — Queries

15 marks

Ponder Stibbons needs several reports generated for the Faculty Meeting.

2a) SQL: Find long modules

(5 marks)

Using SQL, retrieve all modules where `duration_hrs` is greater than 60. Display the `module_name` and `duration_hrs` columns only.

2b) pandas: Filter female students

(5 marks)

Load the **students** table into a DataFrame, then use pandas boolean indexing to find all students where `gender` is 'F'.

2c) SQL: Sort teachers by name

(5 marks)

Using SQL, display all teachers sorted alphabetically by name in ascending order.

Task 3 — Data Entry Form

10 marks

New students keep arriving at the Registry office (several are noticeably on fire, which Ridcully considers a good sign). Create an ipywidgets form to register a new student.

Your form must include:

- A Text widget for the student's **name**
- A Text widget for their **date of birth** (format: YYYY-MM-DD)
- A Dropdown for **gender** — options: F, M, Other
- A Dropdown for **programme**, populated from the programs table
- A **Submit** button with `button_style='success'`

When submitted, the form must:

- INSERT the new student into the **students** table and commit
- Display a confirmation message with the student's name and programme
- Clear the name and DOB fields ready for the next entry

Marking breakdown	Marks
Widgets created correctly	4
INSERT executes and commits	3
Confirmation message displayed	2
Form displayed and functional	1

Task 4 — Report and Visualisation

10 marks

The Bursar has requested a visual summary for the next Senate meeting (he will not read prose under any circumstances).

4a) Query: Students per programme

(4 marks)

Write a SQL query using **JOIN** and **GROUP BY** to count how many students are enrolled in each programme. Your result must show the **programme name** and a **count** column.

Hint: join students and programs on program_id.

4b) Chart: Bar chart of students per programme

(6 marks)

Using the result from Task 4a, create a bar chart showing the number of students in each programme. Your chart must include a title, axis labels, and `plt.tight_layout()`.

Task 5 — Import External Data

5 marks

A batch of late-enrolling students has arrived by carrier pigeon from Überwald. Their records are in a CSV file called `late_enrolments.csv`, created by running the first cell in Task 5.

Write code to:

1. Read the CSV into a pandas DataFrame
2. Append those records to the **students** table using `to_sql`
3. Verify the import by displaying the total student count after the import

Before submitting

- Run **Kernel > Restart & Run All** to confirm every cell executes without errors.
- Check all outputs are visible in your notebook.
- Save your file one final time with Ctrl+S.
- Upload your `.ipynb` file to Moodle.

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The Librarian would like it noted that all library books must be returned before results are released. Ook.